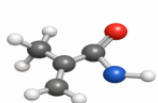


1 Inputs and dependencies



Molecule



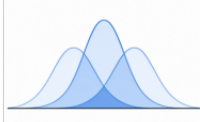
geometry

Charge



closed shell

Basis set



STO-3G, 6-31G(d)

PyQInt



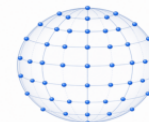
analytic integrals

NumPy/SciPy



arrays, harmonics

PyLebedev



angular quadrature

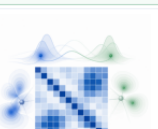
2 Core PyDFT construction



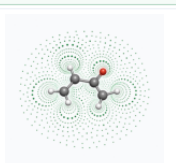
Basis amplitudes



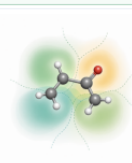
$S_{\mu\nu}, T_{\mu\nu}, V_{\mu\nu}$



Becke grid
 $\{\mathbf{r}_i, w_i\}$



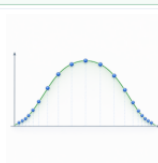
Fuzzy weights
 $w_A(\mathbf{r})$



Lebedev
 $\int d\Omega$



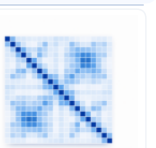
Radial grid
 r_k



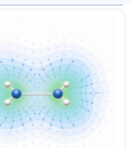
3 Kohn-Sham iteration



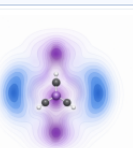
Density matrix
 $P_{\mu\nu}$



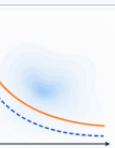
$\rho(\mathbf{r})$
 $\nabla\rho(\mathbf{r})$



Hartree
 $v_H(\mathbf{r})$



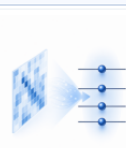
XC
 $V_{xc}^{LDA/PBE}$



Build Fock
 $F = H + J + V_{xc}$



Solve KS
 $FC = SC\epsilon$



Update density
 $P = 2CC^T$



Converged?
 $|\Delta E|, |\Delta P| < \tau$

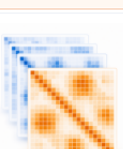
Yes

No

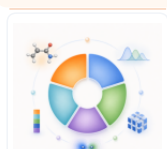
4 Exposed learning outputs



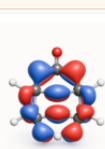
Matrices
 F, P, S



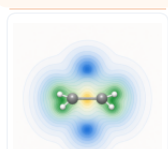
Energy terms
 E_i



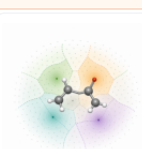
Orbitals
 ψ_i



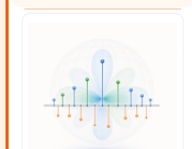
Density maps
 $\rho(\mathbf{r})$



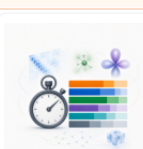
Becke cells
 $w_A(\mathbf{r})$



Spherical coeff.
 ρ_{klm}



Timing breakdown



Interactive exploration

